

RORBACKSNAS, Sweden

WMO: 023060

Lat: 61.133N

Lon: 12.817E

Elev: 1526

StdP: 13.90

Time Zone: 1.00 (EUC)

Period: 82-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-12.4	-21.8	1.7	-17.9	-16.2	2.4	-12.4	23.2	29.2	22.4	30.2	0.5	0	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	77.5	60.5	73.8	58.4	69.9	56.2	61.4	74.9	59.4	71.3	57.5	67.6	4.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.1	70.9	62.7	54.3	66.3	61.2	52.7	62.5	60.5	28.0	75.0	26.6	71.8	25.4	68.0	67.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	16.0	14.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.2	15.7	18.5	24.9	33.2	45.0	51.6	56.3	52.6	44.9	36.2	25.2	17.8		
	DBStd	16.59	13.93	11.21	8.65	7.02	6.51	6.15	6.09	5.92	5.71	6.97	9.50	12.31		
	HDD50	5856	1064	882	778	505	186	51	12	36	169	428	745	1000		
	HDD65	10871	1529	1302	1243	954	620	402	278	388	602	893	1195	1465		
	CDD50	470	0	0	0	0	31	100	206	116	17	0	0	0		
	CDD65	11	0	0	0	0	0	1	7	3	0	0	0	0		
	CDH74	280	0	0	0	0	2	41	158	79	0	0	0	0		
	CDH80	29	0	0	0	0	0	1	11	17	0	0	0	0		
	WSAvg	4.2	4.5	3.9	4.5	4.5	4.5	4.7	4.1	3.6	4.2	4.3	3.5	3.6		
	PrecAvg	30.7	2.0	1.5	1.3	1.7	2.5	3.3	3.5	4.2	3.0	3.0	2.6	2.1		
	PrecMax	42.9	4.6	2.5	2.8	4.1	6.7	6.1	6.7	6.5	5.2	6.3	7.3	4.0		
(15) Precipitation	PrecMin	24.3	0.4	0.3	0.2	0.4	0.4	0.3	0.8	1.7	1.0	0.5	0.9	0.2		
	PrecStd	3.7	1.0	0.6	0.6	1.1	1.4	1.4	1.2	1.4	1.1	1.4	1.5	0.9		
	DB	39.2	42.2	45.2	61.1	73.9	79.9	81.8	81.7	63.0	51.7	47.0	38.8			
	MCWB	36.1	38.2	37.8	48.6	56.1	60.4	62.5	62.6	53.5	47.5	45.2	36.6			
	2% DB	36.7	39.4	39.9	53.1	69.7	74.5	78.7	74.6	59.8	49.8	42.5	37.1			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	34.1	36.0	34.1	43.2	53.4	58.9	61.3	59.1	52.2	46.0	40.3	35.4			
	5% DB	34.4	35.6	37.3	48.5	64.5	71.0	75.5	68.6	56.6	48.0	40.2	35.0			
	MCWB	32.0	33.4	33.6	40.1	51.0	56.5	59.4	56.9	50.9	44.5	38.3	33.8			
	10% DB	32.4	33.2	35.3	44.9	60.0	66.7	70.9	63.7	54.2	46.0	37.2	32.9			
	MCWB	30.7	31.7	32.6	37.7	48.8	54.0	57.3	54.0	48.8	43.3	35.7	31.7			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	36.9	39.5	38.6	48.9	57.5	61.8	64.1	62.7	57.2	50.0	45.4	37.6			
	MCDB	38.3	41.6	42.6	60.4	70.4	77.3	76.4	78.3	60.5	51.1	47.0	38.4			
	2% WB	34.6	36.5	36.0	44.1	55.2	59.4	62.0	60.0	54.6	47.2	40.6	35.8			
	MCDB	36.1	38.5	38.0	52.2	65.7	72.1	75.8	71.5	57.6	48.6	41.9	36.6			
	5% WB	32.5	33.9	34.4	41.1	52.5	57.7	60.4	57.8	51.8	45.3	38.5	33.8			
(31) Mean Daily Temperature Range	MCDB	33.9	35.2	36.5	46.7	62.8	69.4	73.1	66.6	55.3	47.0	39.9	34.9			
	10% WB	30.6	31.9	32.8	38.9	49.8	55.3	58.3	55.8	49.5	43.3	35.9	31.8			
	MCDB	32.1	33.1	34.9	43.0	59.0	64.9	69.6	61.9	52.8	45.5	37.2	32.8			
	5% DB	12.9	15.9	15.1	15.1	19.3	19.1	19.4	16.6	15.0	11.2	11.0	12.6			
	MCDBR	10.4	10.4	14.5	22.1	29.5	29.5	28.0	27.1	19.2	12.6	9.5	11.5			
(36) Clear-Sky Solar Irradiance	MCWBR	9.8	9.0	10.6	13.5	15.0	14.8	13.7	12.9	11.7	9.3	8.7	11.1			
	5% WB	10.5	9.4	11.5	18.5	25.7	25.4	24.7	24.5	13.3	8.2	9.1	10.6			
	MCDBR	10.2	8.8	8.9	11.7	13.3	13.2	12.6	11.8	9.6	7.2	8.5	10.8			
	taub	0.221	0.268	0.296	0.334	0.340	0.333	0.352	0.342	0.312	0.296	0.241	N/A			
	taud	2.156	2.274	2.289	2.351	2.452	2.495	2.453	2.472	2.545	2.495	2.247	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	171	231	263	273	281	284	275	269	259	222	161	N/A			
	Edh at Noon	11	20	28	32	31	30	31	28	22	16	10	N/A			
	RadAvg	90	299	788	1271	1490	1668	1503	1172	769	345	108	50			
(45)	RadStd	12	58	81	136	144	128	203	112	94	62	29	6			

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page